QUESTION NO 01:

The data in the table was obtained by Robert A. Millikan (United States, 1868-1953), who was awarded the 1921 Nobel prize in physics for his verification of the photoelectric effect. Millikan's photoelectric data for lithium are:

Wavelength (nm)	433.9	404.7	365.0	312.5	253.5
Stopping Potential (V)	0.55	0.73	1.09	1.67	2.57

a. Make a plot of the table on graph paper with proper scaling.
b. Find the plank's constant.
c. Work function for lithium.

OBJECT:

To determine the Planck's Constant and Work function.

WORKING FORMULA:

• $c = f\lambda$ • $h = me$ • $\phi = hfo$

CALCULATION:

First, we find out the frequencies to plot the graph;

A) For Frequencies:

$$\because f = \frac{c}{\lambda}$$

1. $f_1 = \frac{3\times10\text{^}8}{433.9\times10\text{^}-9} = 6.91 \times 10^{14}\,Hz$

2. $f_2 = \frac{3\times10\text{^}8}{404.7\times10\text{^}-9} = 7.41 \times 10^{14}\,Hz$

3. $f_3 = \frac{3\times10\text{^}8}{365\times10\text{^}-9} = 8.21 \times 10^{14}\,Hz$

4. $f_4 = \frac{3\times10\text{^}8}{312.5\times10\text{^}-9} = 9.6 \times 10^{14}\,Hz$

5. $f_5 = \frac{3\times10\text{^}8}{253.5\times10\text{^}-9} = 11.8 \times 10^{14}\,Hz$

B) For Plank's Constant:

$\because h = me$

For "m":

$\because m = \frac{y2 - y1}{x2 - x1}$

$\left.\begin{array}{ll} x_1 = 8 \times 10^{14} & , x_2 = 10 \times 10^{14} \\ y_1 = 1 & , y_2 = 1.82 \end{array}\right\}$ FROM GRAPH

$$m = \frac{1.83 - 1}{10 \times 10^{14} - 8 \times 10^{14}}$$

$m = 4.15 \times 10^{-15}$

$\because h = me$

$h = 4.15 \times 10^{-15} \times 1.6 \times 10^{-19}$

$\boxed{h = 6.64 \times 10^{-34}\ \text{Js}}$

C) For Work Function:

$\because \phi = hfo$

$\phi = 6.64 \times 10^{-34} \times 5.41 \times 10^{14}$

$\phi = 3.59 \times 10^{-19}\ J$

$\phi = \frac{3.59 \times 10^{-19}}{1.6 \times 10^{-19}}$

$\boxed{\phi = 2.24\ \text{eV}}$

RESULT:

- The calculated Value of Planck's Constant is $\underline{6.64 \times 10^{-34}\ \text{Js.}}$
- The calculated Work Function of Lithium is $\underline{2.24 \times 10^{-19}\ eV.}$

QUESTION NO 02:

Sodium and silver have work function of 2.64eV and 4.73eV, respectively.

a. If the surfaces of both metals are illuminated with same monochromatic light which metal will give off photons with greater speed? How much faster will those electrons be?

b. Calculate the cutoff wavelength of both metals?

Given Data for both parts :

ΦNa (sodium's work function) = 2.64 eV

ΦAg (silver's work function) = 4.73 eV

a)

Solution:

The energy of incident light E is assumed to be constant for both the metals. The K.E of emitted electrons is given by:

$K.E = hf - \Phi$

$K.E = E - \Phi$

As we know that, $K.E = \frac{1}{2} mV^2$

Then,

$$V = \sqrt{2K.E/m}$$

The ratios of K.E can be given as:

K.E of Na= $E - \Phi Na$

K.E of Ag= $E - \Phi Ag$

To eliminate E,

Since $\Phi Na < \Phi Ag$ then, K.E of Na > K.E of Ag.

$\Delta V = V_{Na} - V_{Ag}$

$\Delta V = \sqrt{2/m}\,(\sqrt{E - \Phi Na} - \sqrt{E - \Phi Ag})$ $\longrightarrow$ (After cancelling E)

$\Delta V = \sqrt{2/m}\,(\Phi Na - \Phi Ag)$

Substitute the values:

$\Delta V = \sqrt{2/9.1*10^{-31}} *(\sqrt{4.73-2.64})$

$\Delta V = 8.57x10^5$ m/s

RESULT:

- This proves that the electrons emitted from Sodium are 8.57x10^5 m/s faster than that of Silver, the reason being the work function (Φ) of Sodium (Na) is lower than that of Silver (Ag).

b)

Solution:

Cut off wavelength is given by:

$\lambda Na = hc / \Phi$ $\longrightarrow$ (1)

$\lambda Ag = hc / \Phi$ $\longrightarrow$ (2)

from eq.(1)

$\lambda Na = hc / \Phi$

$\lambda Na = (6.625x10\text{-}34)x(3x108)/(4.23x10\text{-}19)$

$\lambda Na = 4.7x10\text{-}7m$ or 470 nm

from eq.(2)

$\lambda Ag = hc / \Phi$

λAg = (6.625x10-34)x(3x108)/(7.58x10-19)

λAg= 2.62x10-7m or 262 nm

RESULT:

- cutoff wavelength of sodium (Na) : 470nm
- cutoff wavelength of silver (Ag) :262nm